# ELSTE Data science

*Class 1: Introduction to data analysis and inference using Python*


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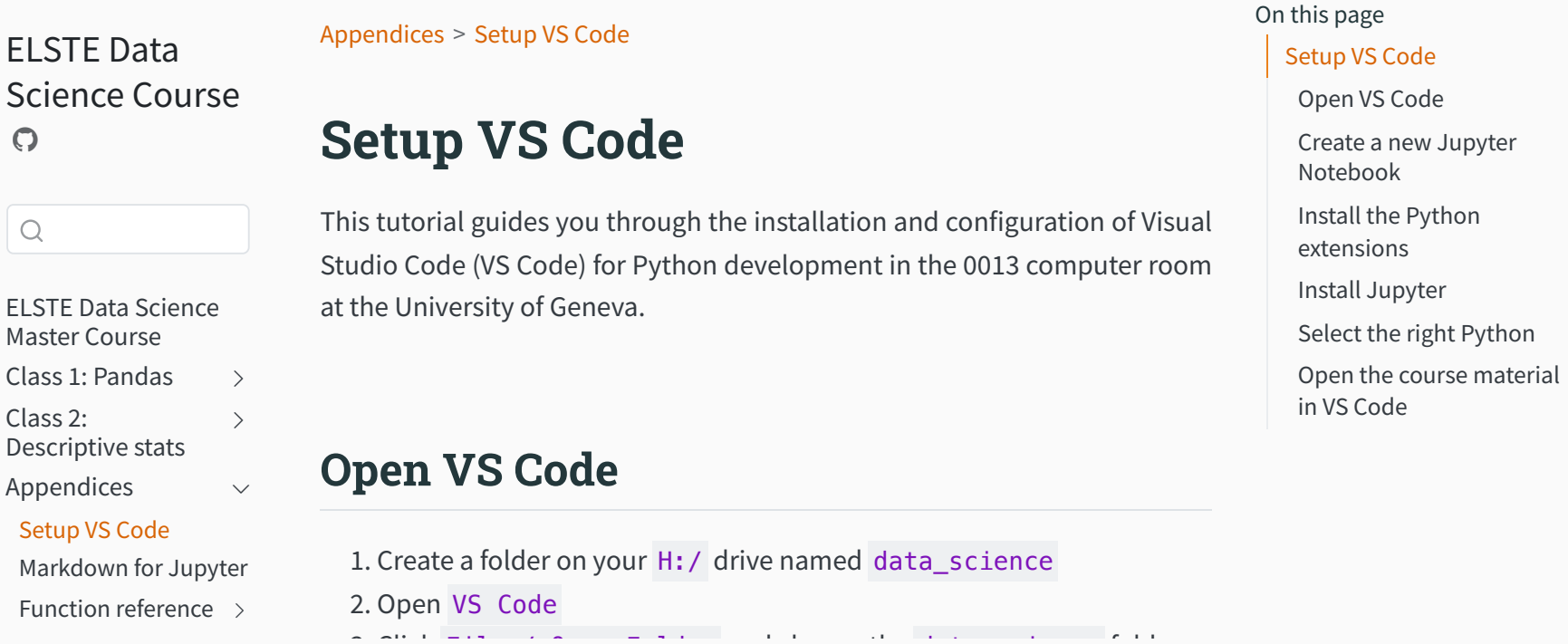
*Earth Sciences*

*Pharmaceutical Sciences*


October 14, 2026

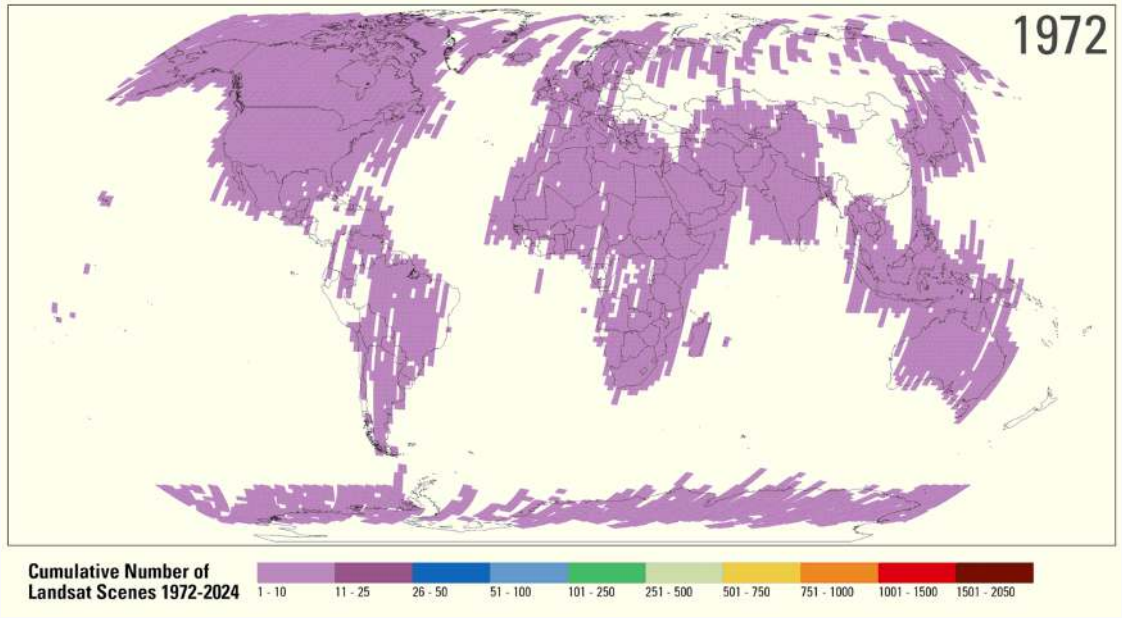
# Get ready

- Go to https://elste-master.github.io/Data-Science/
  - → **Download slides:** Class 1 > Overview
  - → **Setup VS Code:** Appendices > Setup VS Code



# Why this class? (1/2)

*Data deluge: The big EO data landscape*




- **Landsat:**
  - → Download: 53 images/day (2001) - 220’000 images/day (2017)
  - → 5 million images of the Earth surface → > 5 PB
- **ESA Sentinel 1 & 2 →** 4.6TB /daily!

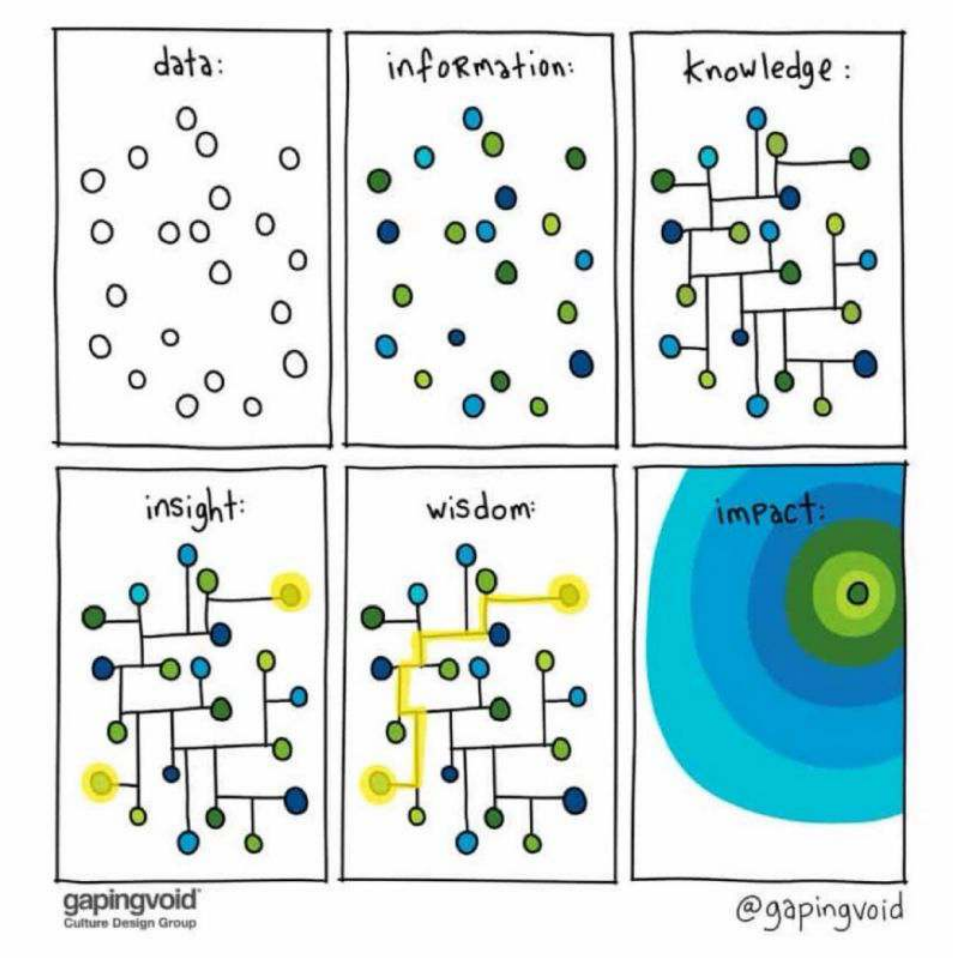
# Why this class? (2/2)

**As scientists, we are exposed to**:

- Increasing computational power/facilities
- Increasing amount of data

**How do we make sense of it all?**

- Scientific coding → gateway to scientific data analysis
- Data science → extracting meaningful infomation and insights from data



# What is data science

*Wikipedia*

> *Data science is an interdisciplinary academic field that uses statistics, scientific computing, scientific methods, processing, scientific visualization, algorithms and systems to extract or extrapolate knowledge from [...] data.*

*National Institutes of Standards and Technology*

> *The field that combines domain expertise, programming skills, and knowledge of mathematics and statistics to extract meaningful insights from data.*

# Class schedule

- **Fall part at UNIGE**: 6 × 3-h long sessions
- **Assistants**: Sofia Hernandez Castro, Henry Hoult
- **Format**: Theory, live coding and exercises

*October 14 and 21*

- **Seb** → Exploratory data analysis

*October 28, November 4 and 11*

- **Stéphane** → Inference

*November 18*

- Question session + exam

# Class objectives

- Scientific coding and stats often appear daunting
- True, there can be a steep learning curve

*Objectives*

- **Not** to make you and expert developer / statistician **but**
  - → Motivate scientific reasoning
  - → Develop statistical thinking

# Get involved!!!

## We want this class to be useful for your research!

Try to contextualise the course material to **your research**:

- Do you already have your own datasets you could bring to class?
- Can you use the course to formulate new research questions?
- Do you know any open-access datasets relevant to your fields?

  → If not, can you find some for the next weeks?

# Why coding?

# Motivations (1/6)

- **Example**: A common - but unnecessarily complicated - workflow of many specialised softwares




- **Objective**: Integrate full research workflow in an environment supporting all tasks



# Motivations (2/6)

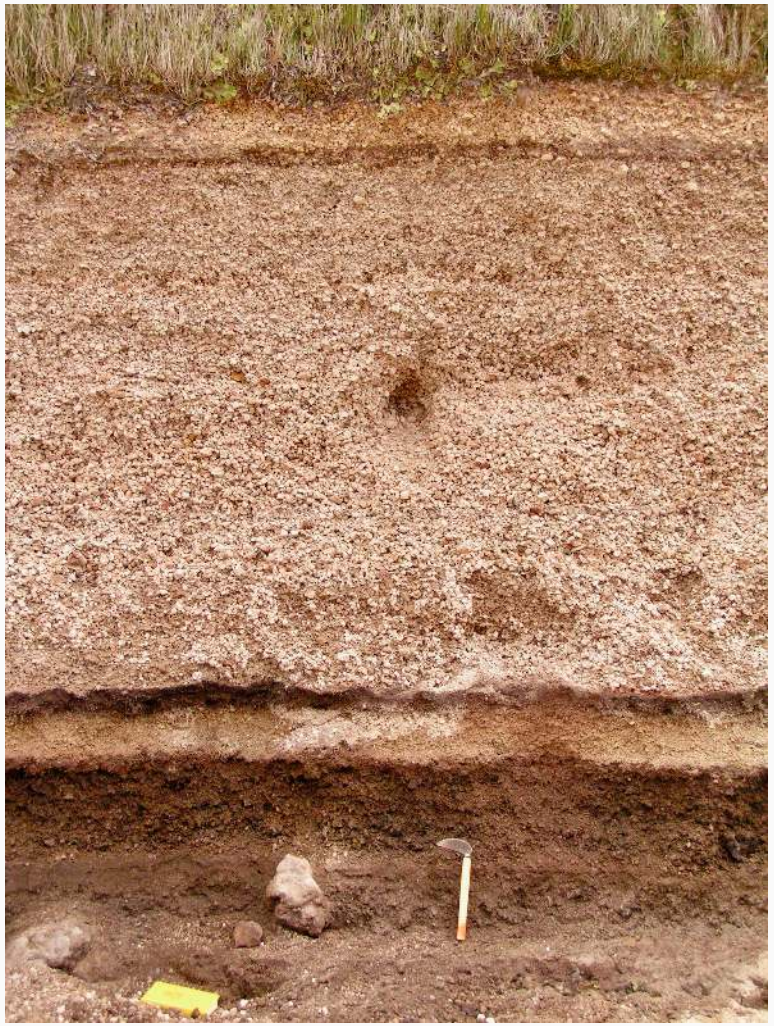
*Motivation 1: Automation*

## Cotopaxi volcano

- Reconstructing eruption source parameters (ESP) from tephra deposits

  → *How do different measurement methods influence ESP estimates?*

- 40 outcrops
- ~10 samples per outcrop
- ~7 measurement methods per sample



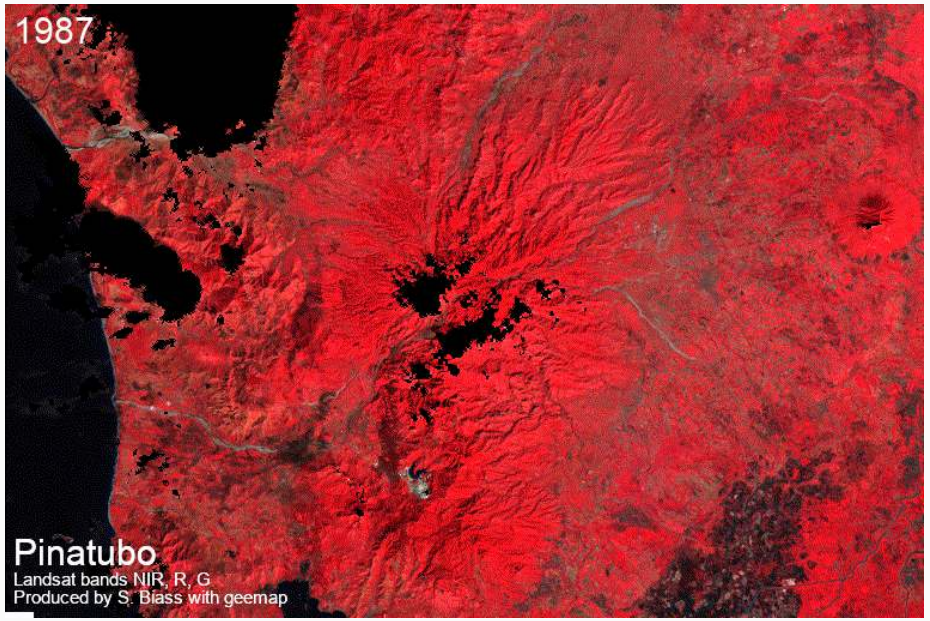
# Motivations (3/6)

*Motivation 2: Data analysis*

**Example 1**: Exploit catalogues of big Earth Observation data

- **Revisit big EO catalogues to infer new knowledge**

  → *What controls the impact and recovery of vegetation following eruptions?*

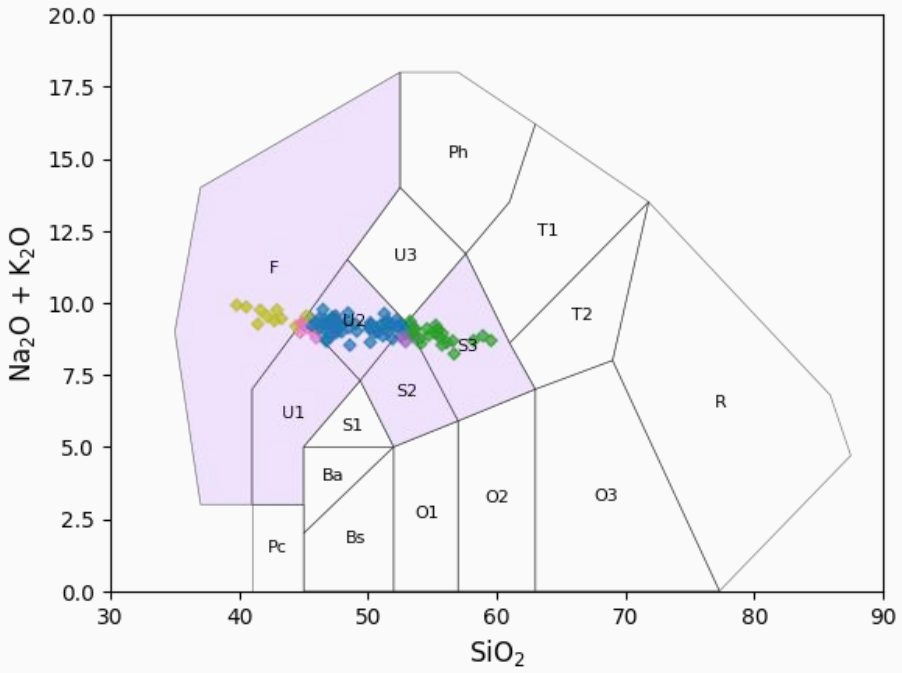


# Motivations (4/6)

*Motivation 2: Data analysis*

**Example 2**: Streamline global geochemical analyses

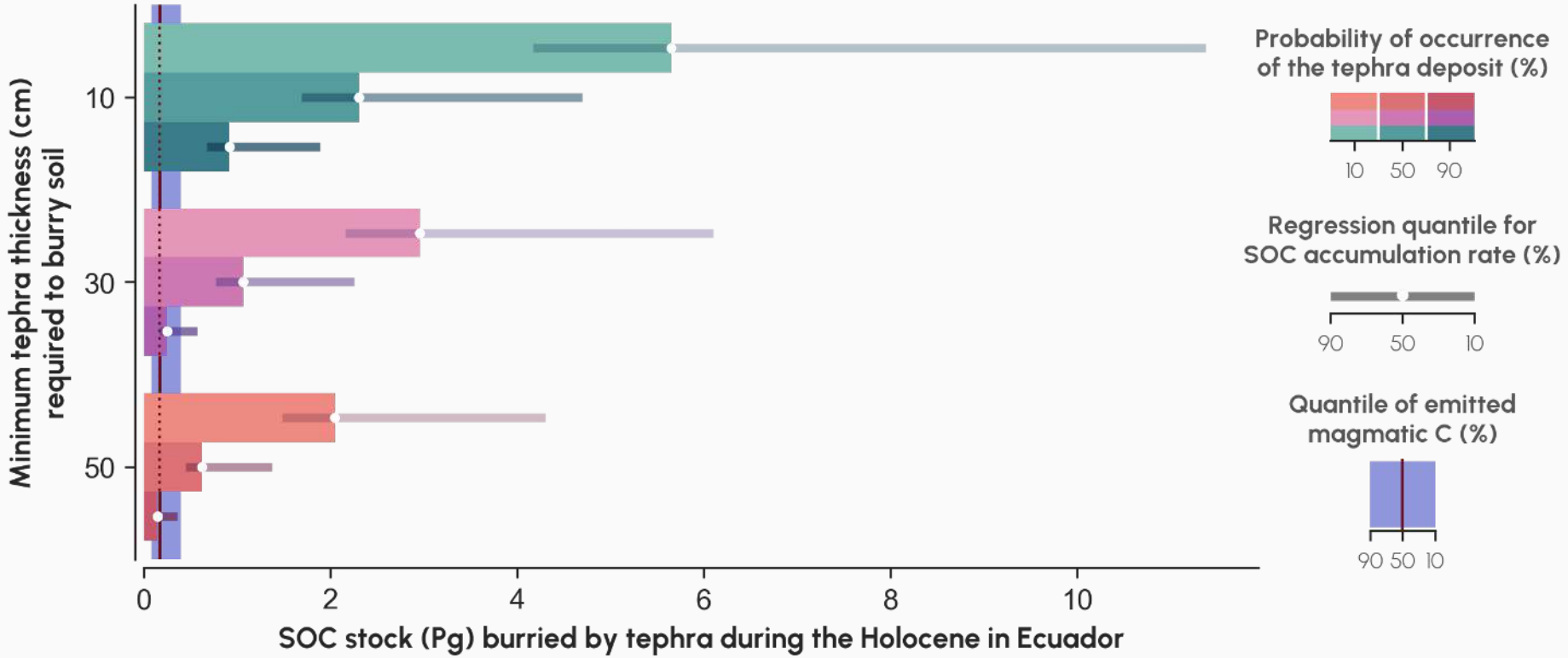
- **Access global databases**
  - → e.g., *Georock* database
- **Automatic dedicated analyses/plots**
  - → e.g., *TAS* diagrams and classification using `pyrolite`



# Motivations (5/6)

*Motivation 3: Visualisation*

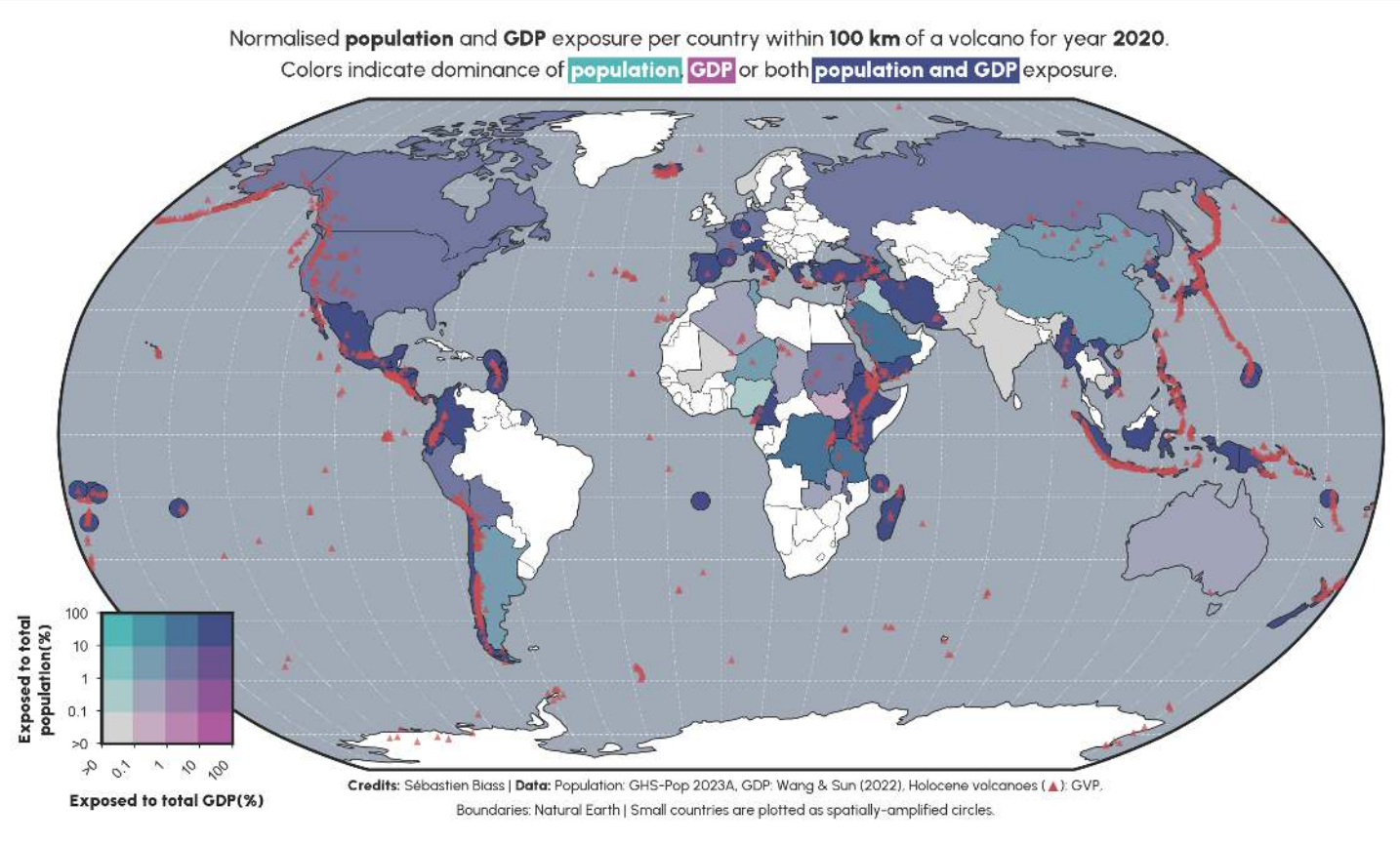
**Example 1**: In volcanically-active regions, soil burial after explosive eruptions capture more carbon that they emit.



# Motivations (6/6)

*Motivation 3: Visualisation*

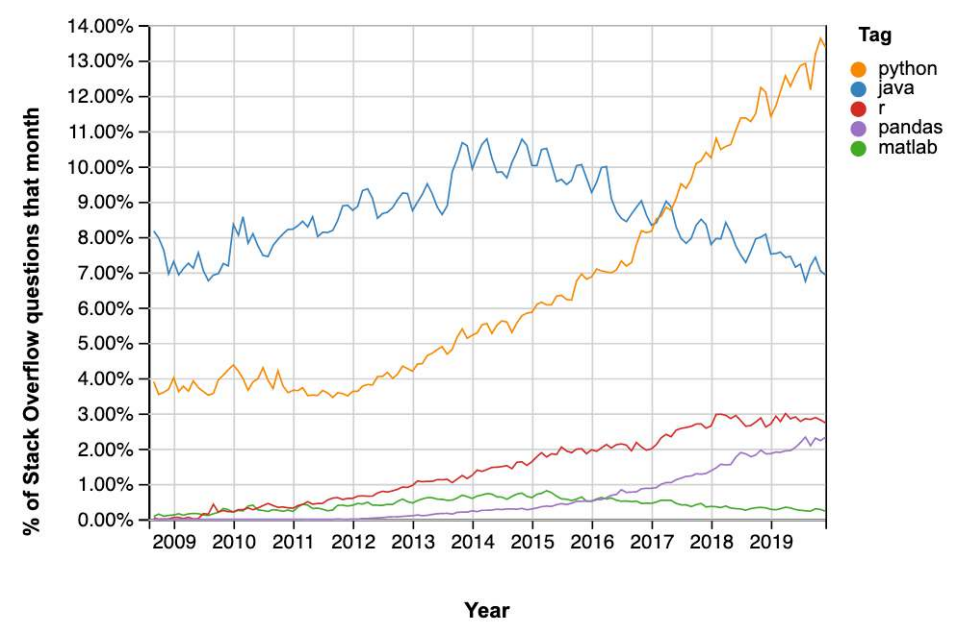
**Example 2**: In Small Island States, ≥90% of population and GDP is exposed to volcanic hazards.



# Why Python?

# Python's advantages [1]

1. **Free**
2. **Open source**
   - Not dependent on any company
   - Large online community
3. **Old** → stable (started in 1991)
4. **Very** popular in science
5. Relatively easy to learn




1. Inspired by the School of Oceanography at University of Washington

# What is Python

## High-level computing language

- Compiled → doesn't directly speak to the computer, but is *interpreted* by another language
  - → Usually slower, but much, much easier to use!

## Modular

- Python is composed of a *core* of functions complemented by wide ecosystem of specialized packages for specific tasks
  - → A function is a bit of code to perform a **specific task**
  - → A module is a collection of **functions**
  - → A package is composed by multiple **modules**

# An example of packages, modules and functions

In the code below, we first load the necessary packages and libraries:

- `matplotlib` is the main **visualisation** package used in Python
- `pyplot` is a module of `matplotlib` that provides easy-to-use functions for **plotting** data
- `figure` is a function of `pyplot` and is the main function to prepare a plot

```
1 # Import the packages and modules
2 from matplotlib import pyplot as plt
3 # Set up a figure for plotting
4 plt.figure(...)
```

# How to use Python

**Option 1**: Your own computer

- Requires some setup
- No internet connection required
- Slow if computer is slow
- All data stored on your own computer

> Try this tutorial

**Option 2**: In the cloud → Google Collab

- No setup required
- No internet connection required
- Decent speed, free as long as Google says it is free
- All data stored in the cloud

# How to run Python

## 3 main ways to run Python…

1. Run Python `.py` scripts from the command line → *deprecated*
2. Run blocks of code from within a Python `.py` script using `ipykernel` → **similar behaviour to R/Matlab**
3. Use **Jupyter Notebooks** in `.ipynb` files

## …in two different environments

1. Dedicated software: VSCode, PyCharm, Spyder
2. Web browser: JupyterLab

# How to approach coding

# Python as a natural language

| Natural language | Python | Example |
|---|---|---|
| Noun | Object or variable | `df` |
| Verb | Function or method | `mean()`, `sort_values()` |
| Complement | Target variable | `"altitude"` |
| Adverb | Argument specifying how | `ascending=False` |
| Sentence | Instruction | `df.sort_values(...)` |

```
1 df.sort_values("altitude", ascending=False)
```



Python checks the grammar of your code. **You** must check the scientific meaning of the sentence.

# From a sentence to a sequence of tasks

| Task | Questions to ask |
|---|---|
| **Load** | Where are the data? How do I read them? |
| **Inspect** | What do the rows, columns and values represent? |
| **Select / filter** | Which columns and observations do I need? |
| **Transform** | What should I sort, recode or calculate? |
| **Summarise / analyse** | What quantity or relationship answers my question? |
| **Visualise / export** | How should I show or save the result? |

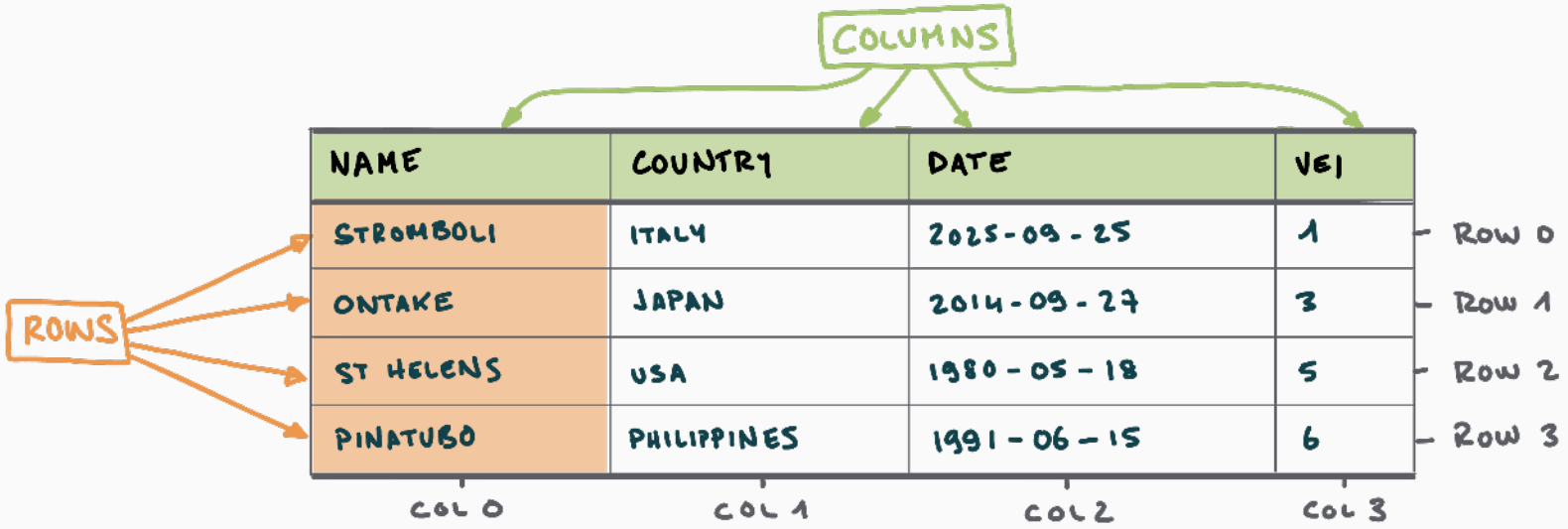
# Today’s plans

# Pandas

`pandas`: A **package** for data manipulation and analysis handling **structured data**

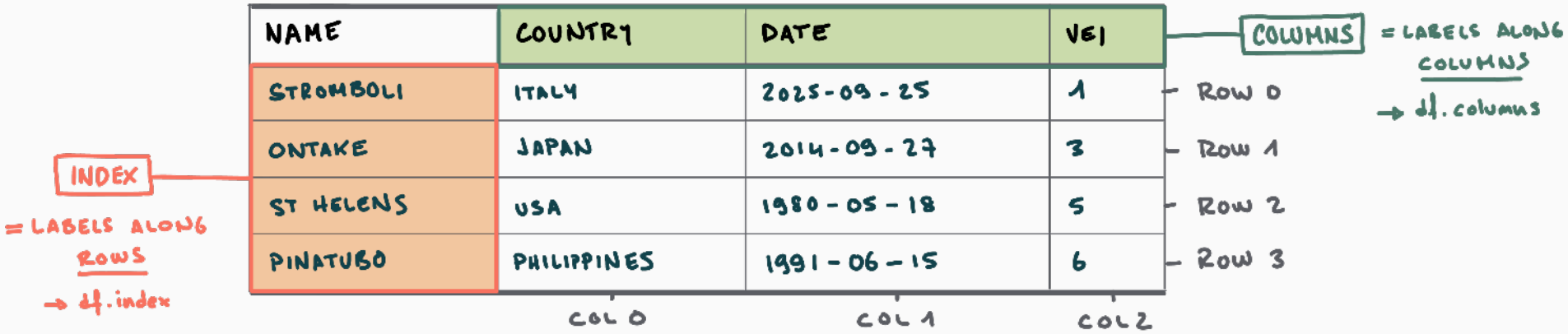
- **Reading/writing data** from common formats (CSV, Excel, JSON, etc.)
- Handling **missing data**
- **Filtering**, **sorting**, **reshaping** and **grouping** data
- **Aggregating** data (sum, mean, count, etc.)
- **Time series support** (date ranges, frequency conversions)
- **Statistical operations**

# Anatomy of a DataFrame (1/2)




- Similar to *Excel* → contains *tabular data* composed of rows and columns
- In *Excel*:
  - → **Rows** are accessed using *numbers*
  - → **Columns** are accessed using *letters*

# Anatomy of a DataFrame (2/2)



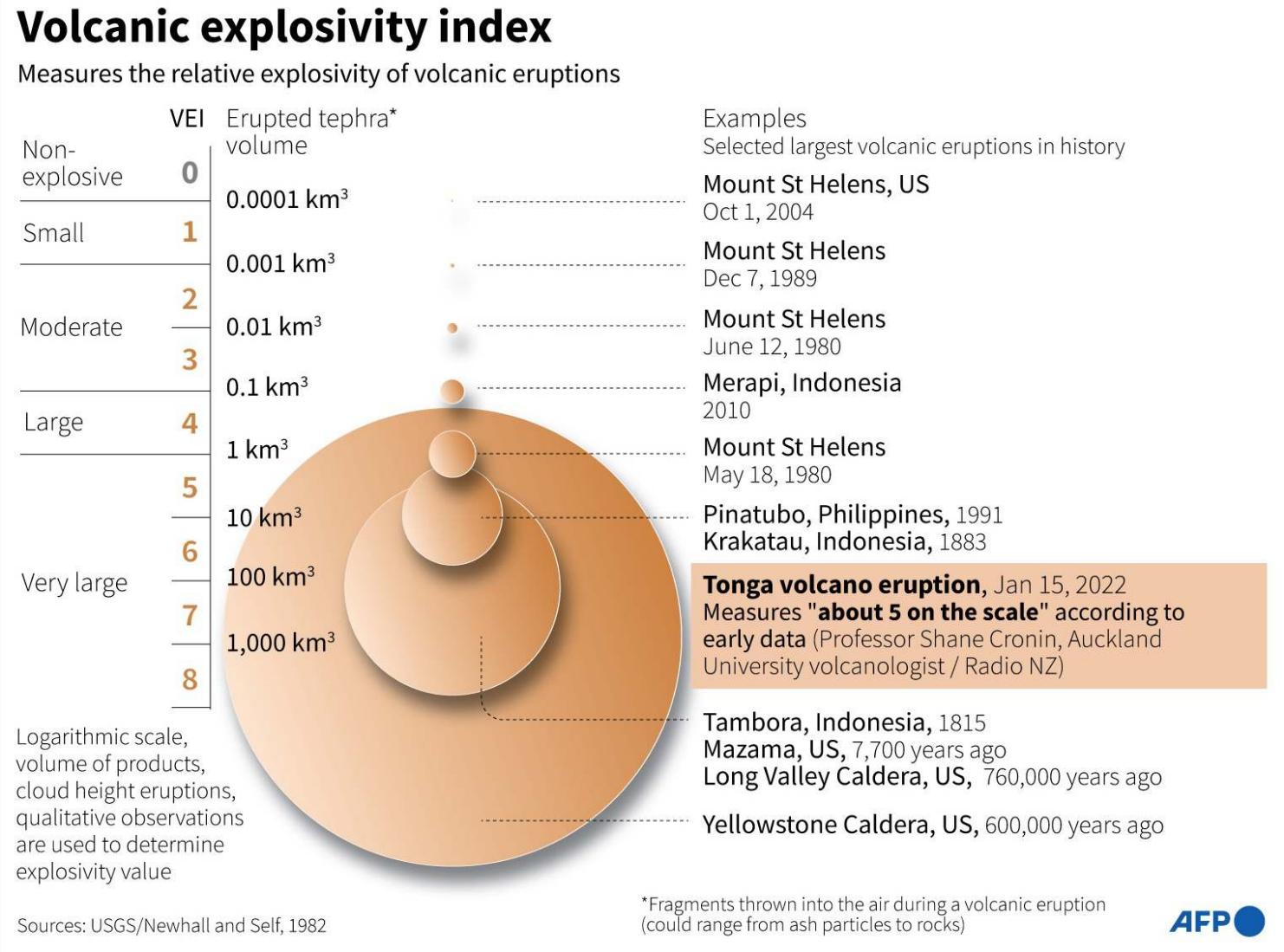

- Unlike *Excel*, **rows** and **columns** can be labelled
  - → Index refers to the label of the **rows**. In the *index*, **values are usually unique** - meaning that each entry has a different label.
  - → Column refers to the label of - logically - the **columns**

# Today's dataset

Synthetic dataset of **selected volcanic eruptions** → first 5 rows:

| Name | Country | Date | VEI | Latitude | Longitude |
|---|---|---|---|---|---|
| St. Helens | USA | 1980-05-18 | 5 | 46.1914 | -122.196 |
| Pinatubo | Philippines | 1991-04-02 | 6 | 15.1501 | 120.347 |
| El Chichón | Mexico | 1982-03-28 | 5 | 17.3559 | -93.2233 |
| Galunggung | Indonesia | 1982-04-05 | 4 | -7.2567 | 108.077 |
| Nevado del Ruiz | Colombia | 1985-11-13 | 3 | 4.895 | -75.322 |

# Volcanic explosivity index (VEI)



# Today's objectives

**5 modules**:

1. How to load data in a DataFrame
2. How to access data → *query by label/position*
3. How to filter data → *comparison and logical operators*
4. How to rearrange data → *sorting values*
5. How to operate on data → *arithmetic and string operations*

- We expect you to know how to get started with Python / Jupyter notebook
- Let us know if you are lost or if you already know all of this

  → Let's make it useful for **your research**

# Get started

- Go to https://elste-master.github.io/Data-Science/
  - → **Setup VS Code:** Appendices > Setup VS Code
  - → Follow Class 1

**How to do it**?

- Create a Jupyter Notebook
- Copy & paste code fragment → make sure you **understand** what you are doing!
  - → Use the documentation / **cheat sheets** / **companion course**
- Document your code with **Markdown**